Universal gravitational energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the zero-at-infinity convention for gravitational potential energy.

SUCCESS CRITERIA - I can...

1. Use center-to-center distance.
2. Explain why negative potential energy does not mean negative kinetic energy.
3. Compute changes from final minus initial energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Net force rises linearly from zero to 10 N over 0.3 m. Find work and final speed for a 0.3 kg cart starting at rest.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $U = -GMm/r$; $\Delta U = U_f - U_i$.
Outside a spherical source; choose $U = 0$ at infinity. Moving outward makes U less negative.

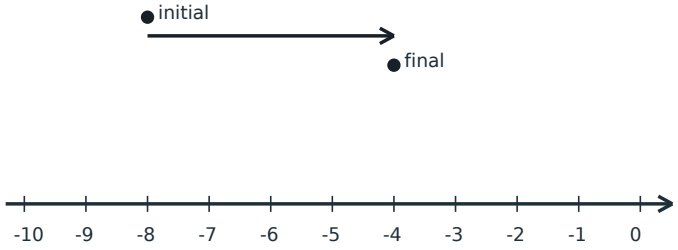
Worked reasoning: For a supplied source with $GMm = 120\ J\cdot m$, potential energy is $U = -120/r$. At $r = 3$ m, $U = -40$ J; at 6 m, $U = -20$ J. Moving outward raises U by 20 J.

Your turn: Draw an energy number line and mark the direction toward the zero at infinite separation.

Marzano check - circle: ① ② ③

03 Represent and interpret

An object moves outward from r to 2 r. Does gravitational potential energy become more or less negative? Draw a number-line comparison.



Example: potential energy (J)

Marzano check - circle: ① ② ③

04 Solve and justify

A 1000 kg satellite moves from $r = 7.0\times10^6$ to 1.4×10^7 m. Use $GM = 4.0\times10^{14}\ m^3/s^2$. Find U at each point andΔU.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A student says “negative potential energy means the object has negative kinetic energy.” Correct the claim.

Marzano check - circle: 1 2 3

06 Choose a method

A toy target needs a 4 km sideways shift over 6×10^{6} s. Its mass is 8×10^{8} kg. Estimate Δv and impulse; state the model used.

Marzano check - circle: 1 2 3

07 Exit check

With GMm=60 J·m, move from r=5 to r=15 m. Find work by gravity.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A mass moves from r=R to 3 R. Express ΔU and work by gravity in terms of GMm/R. Compare with reaching infinity.

Marzano check - circle: 1 2 3